

Surajit Tunga

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EDUCATION

Guru Nanak Institute of Technology

B.Tech. in Electronics and Communication Engineering, CGPA: 8.46 (upto 4th semester)

Kolkata, West Bengal

2023 – Present

Balichak BH Institution

Higher Secondary (XII), Percentage: 76.4%

West Bengal

2022 – 2023

Jamna Ramkrishna Vivekananda Pathamandir

Secondary (X), Percentage: 84.8%

West Bengal

2020 – 2021

TECHNICAL SKILLS

Programming: JavaScript, JAVA, Python, C

Web Development: MERN Stack, Tailwind CSS, Django.

Databases: mySQL, MongoDB, SQL.

Tools & Technologies: Git, GitHub, Vs Code, Matlab, Google Colab.

Machine Learning: scikit-learn, NumPy, Pandas, XGBoost, Flask.

Core Competencies: Data Structures & Algorithms, Database Management System (DBMS), Object Oriented Programming (OOPs).

PERSONAL PROJECTS

Urban Expansion and Land Use Analysis Using Satellite Image Processing [Link](#)

Developed an automated system for detecting and visualizing urban growth using multi-temporal Sentinel-2 imagery. Applied NDVI, NDBI, and NDWI indices with Random Forest classification and Web-GIS-based change detection including urban planning, smart city monitoring.

Forest Cover Type Detection [Link](#)

A machine learning project that predicts forest cover types using environmental features. This project implements a Random Forest classifier to classify forest areas into 7 different cover types based on terrain and soil characteristics.

Resume Builder [Link](#)

Built a customizable Resume Builder web app using MERN Stack. Enabled users to input personal and professional details, select templates, and generate professional-looking resumes in PDF format. Supported real-time preview, template switching, and data persistence.

Code Reviewer [Link](#)

A web application that leverages AI to review code snippets, built with the MERN (MongoDB, Express, React, Node.js) stack.

ACHIEVEMENTS

Smart Bengal Hackathon

May 2025

Finalist

- Developed a prototype for Detection of person talking with mobile in no mobile zone, collaborating with a 4-member team.
- Team Lead & Led backend development using Django, sqlite.

IHMMC 2025 (International Level Hardware Model Making Competition)

April 2025

Winner (10th Position)

- Developed a prototype for hand gesture recognition system facilitate communication for mute individuals through gesture-to-speech translation, collaborating with a 6-member team.
- Team Lead & Led model development using python, open-cv, tensorflow, mediapipe.

CERTIFICATIONS

Introduction to Database System, NPTEL (ELITE Certification)

May 2025

Python with Django, ARDENT COMPUTECH PVT LTD

March 2025

PUBLICATIONS

S Tunga, S. Dasgupta, A kar, T Pramanik, N Banerjee, “Dristi-Robust Change Detection, Monitoring, and Alert System on user-defined AOI using Multi-Temporal LISS-4 Satellite Imagery”. *Journal of Geomatics*, Under Review, 2025.

Surajit Tunga, Somerita Das, Sagnik Roy, Dr. Nabaneeta Banerjee, “Energy Optimization Techniques in Next-Generation Communication Networks”, *Int. J. Sci. Inno. Eng.*, vol. 2, no. 6, pp. 824–830, Jun. 2025, doi: [10.70849/IJSCI](https://doi.org/10.70849/IJSCI)

- Co-author in three additional published research papers in related domain.

AREA OF INTEREST

Image Processing, ML, Web-GIS.